

BIOMECHANICS (EPHE 341)

General kinematics

McGinnis Chpt 7 · angular and linear motion together

By the end of this lecture you should be able to...

- **Explain** why the velocity of a segment depends on the frame of reference you measure it from
- **Calculate** the tangential velocity of a segment as $V_{A/B} = \omega r$
- **Solve** for the magnitude and direction of the resultant velocity
- **Write** the general kinematics equation $V_{A/G} = V_{A/B} + V_{B/G}$
- **Resolve** each velocity into x and y components and add them component by component
- **Extend** the equation along a chain of joints, and use it on a real stride

A combination of angular and linear motion

- General kinematics is a **combination of angular and linear motion**
- It combines
 - angular and linear **displacement**
 - angular and linear **velocity**
 - angular and linear **acceleration**

This lecture will help you determine the kinematics of a **segment rotating about a joint that is itself moving through space**.



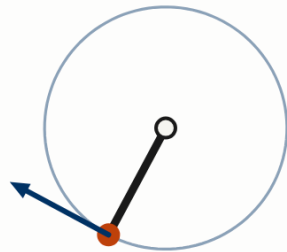
Why the frame of reference matters

- General kinematics demonstrates the importance of **frame of reference** for segmental velocity
- Angular kinematics taught us how to solve for the velocity of a segment rotating about a joint — but **if the joint is moving**, the velocity of the joint will change the velocity of the segment relative to a reference point
- Since there is **both translation and angular motion** in a human system, general kinematics is an important tool



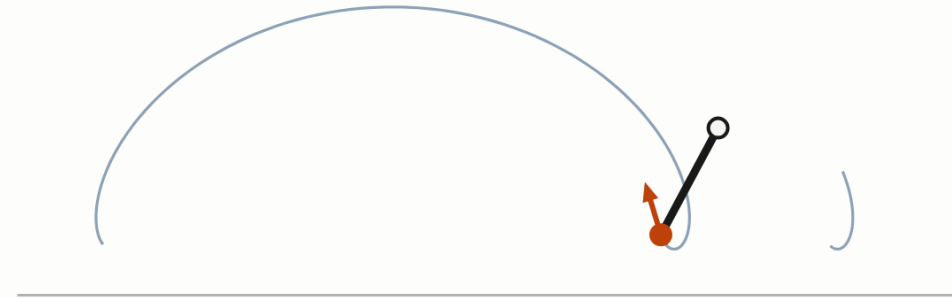
The same segment, seen from two places

Camera on the joint



speed about the joint 2.44 m/s

Camera on the ground



speed over the ground 1.20 m/s

About the joint the hand holds a steady **2.44 m/s** — that is ωr and it never changes. Over the ground it is **3.13 m/s** right now, swinging between **0.64** and **4.24 m/s**.

Angular
velocity ω



4.2
rad/s

Joint
velocity
 $V_{B/G}$

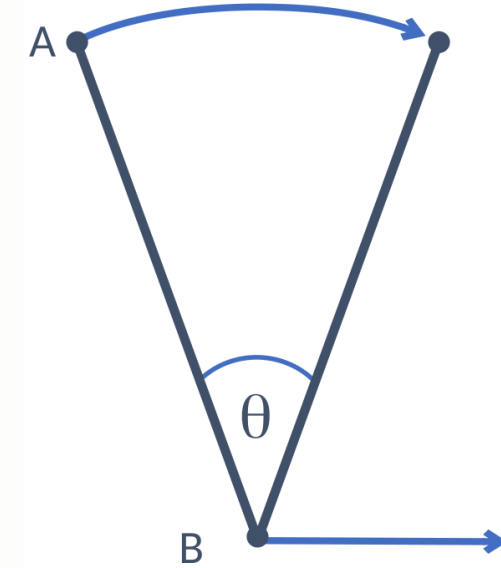


1.8
m/s

|| Pause

What we already know, and what is missing

- From angular kinematics we learned how to determine the velocity of **A relative to B** in both angular and linear terms
- But what if the **whole system was moving**?
- That is, if A is moving relative to B, and B were moving relative to **some other joint or the ground**
- How would we determine the **velocity of A relative to the ground**?

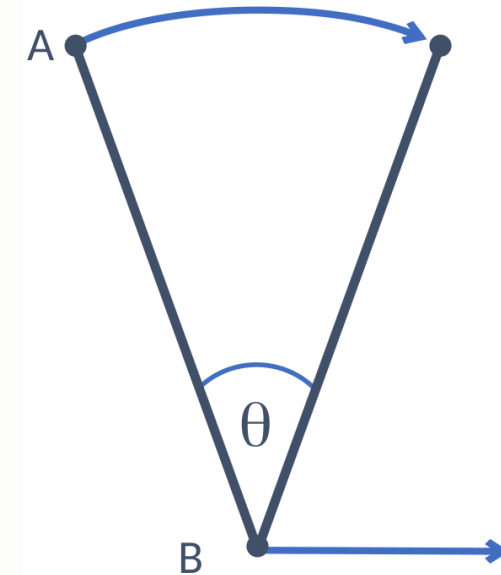


The general kinematics equation

The velocity of a segment relative to the ground ($V_{A/G}$) is the **vector sum** of the segment's velocity relative to the joint ($V_{A/B}$) and the velocity of that joint relative to the ground ($V_{B/G}$).

Written in equation form:

$$V_{A/G} = V_{A/B} + V_{B/G}$$



Where...

$$V_{A/G}$$

is the **linear** velocity of A with respect to the ground

$$V_{A/B}$$

is the **tangential** velocity of A with respect to B

$$V_{B/G}$$

is the **linear** velocity of B with respect to the ground

Only the middle one is **tangential** — it is the term that comes from rotation, and it is always perpendicular to the segment. The other two are ordinary linear velocities.

The tangential term is the one we have to build

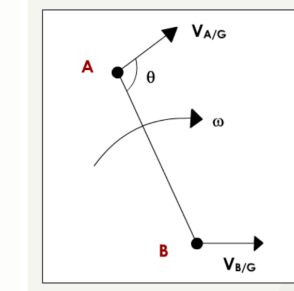
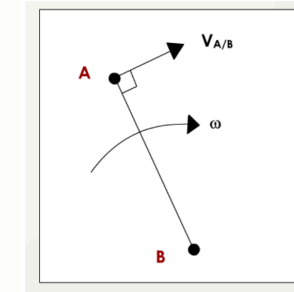
The equation requires the addition of **linear velocity vectors**:

$$\mathbf{V}_{A/G} = \mathbf{V}_{A/B} + \mathbf{V}_{B/G}$$

Assuming that we know $\mathbf{V}_{B/G}$ and the angular velocity of A about B, then we need to determine $\mathbf{V}_{A/B}$, which is the tangential velocity of A about B:

$$\mathbf{V}_{A/B} = \omega \mathbf{r}$$

where ω is the angular velocity of A about B and $\mathbf{r}$ is the radial length from B to A.



One equation for each directional component

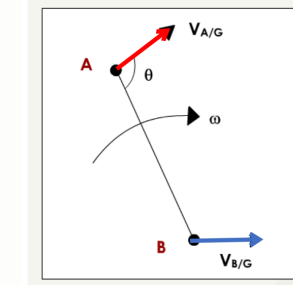
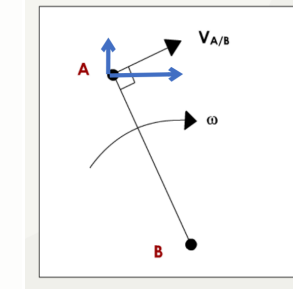
This means that we need an equation for **each directional component**:

$$V_{A/Gx} = V_{A/Bx} + V_{B/Gx}$$

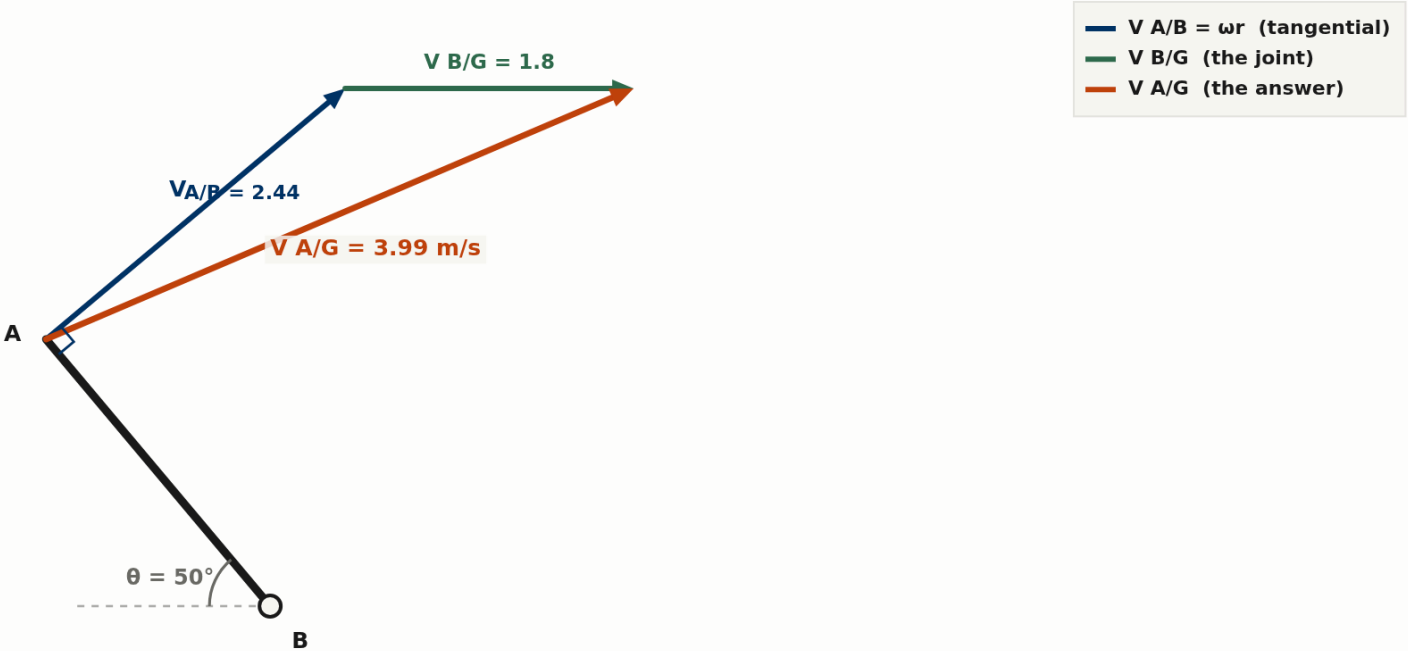
$$V_{A/Gy} = V_{A/By} + V_{B/Gy}$$

Then we need to use **trigonometry** to break up $V_{A/B}$ and $V_{B/G}$ into x and y components, and solve for the resultant velocity $V_{A/G}$:

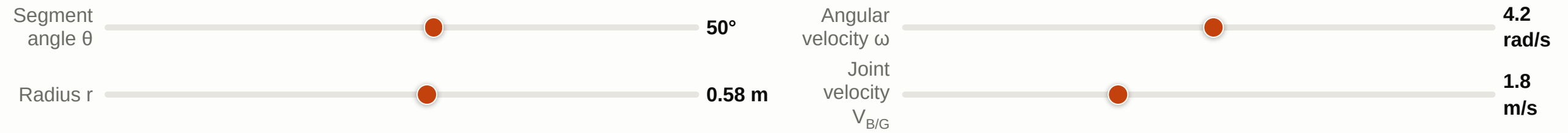
$$V_{A/G}^2 = V_{A/Gx}^2 + V_{A/Gy}^2$$



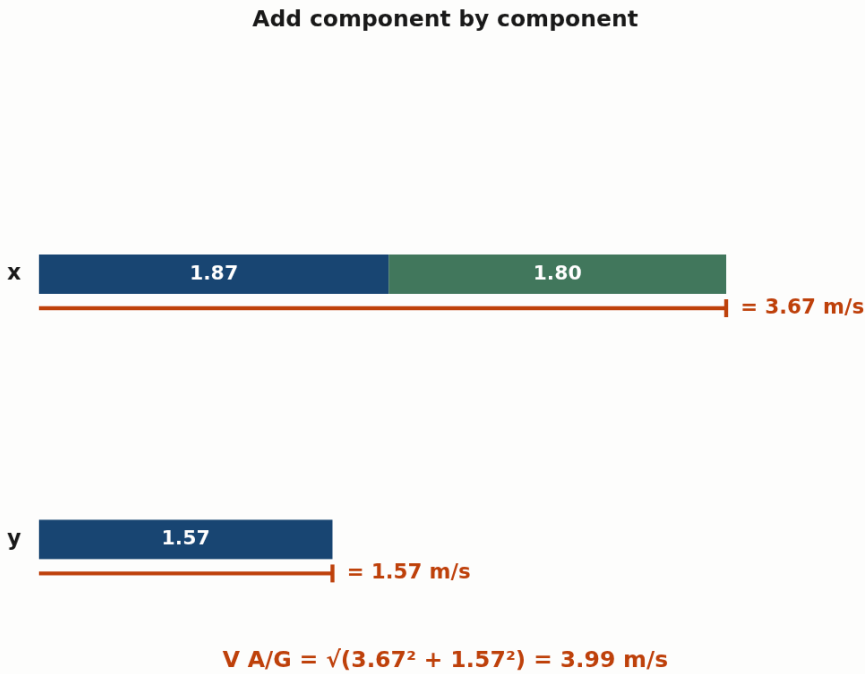
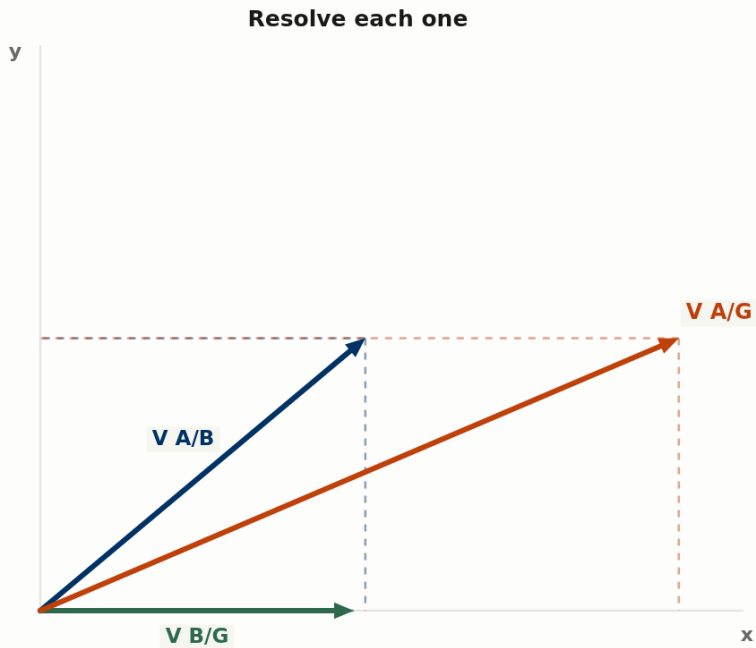
Adding the two vectors



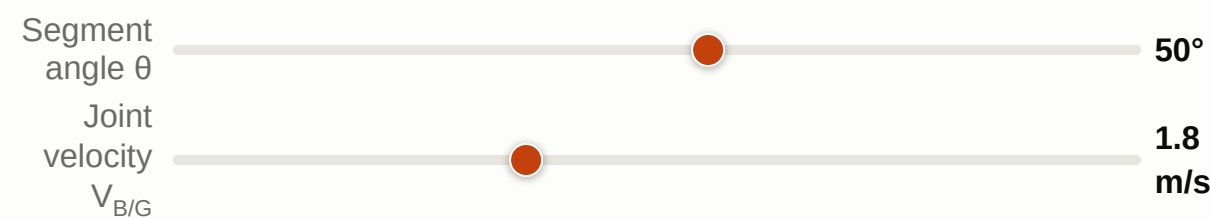
$V_{A/B} = \omega r = 2.44 \text{ m/s}$ perpendicular to the segment, plus $V_{B/G} = 1.8 \text{ m/s}$ horizontal, gives $V_{A/G} = 3.99 \text{ m/s}$ at 23.1° .



Breaking each vector into x and y

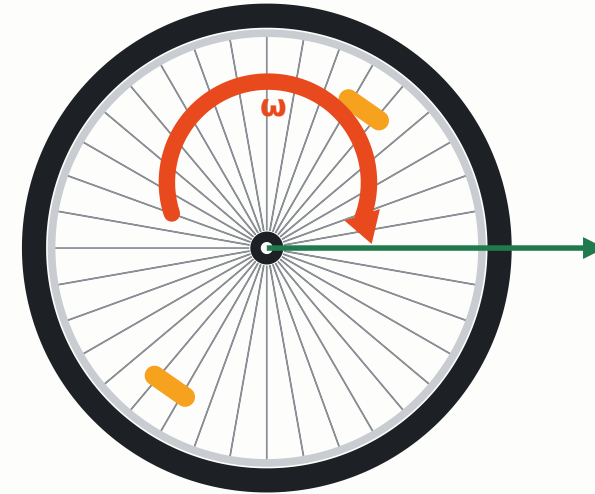


$V_{A/B}$ sits at **40°** to the horizontal — that is $90^\circ - \theta$. Its components are **1.87** and **1.57**; adding $V_{B/G}$ gives **3.67** and **1.57**, so $V_{A/G} = \mathbf{3.99 \text{ m/s}}$ at **23.1°**.



Example Problem

What is the linear velocity of a point on the **top of a bicycle wheel** (radius = 34 cm) if the wheel has an angular velocity of **2 rad/s** and the bicycle has a velocity of **10 m/s**?



Solution

First make an equation for the velocity of A with respect to the ground:

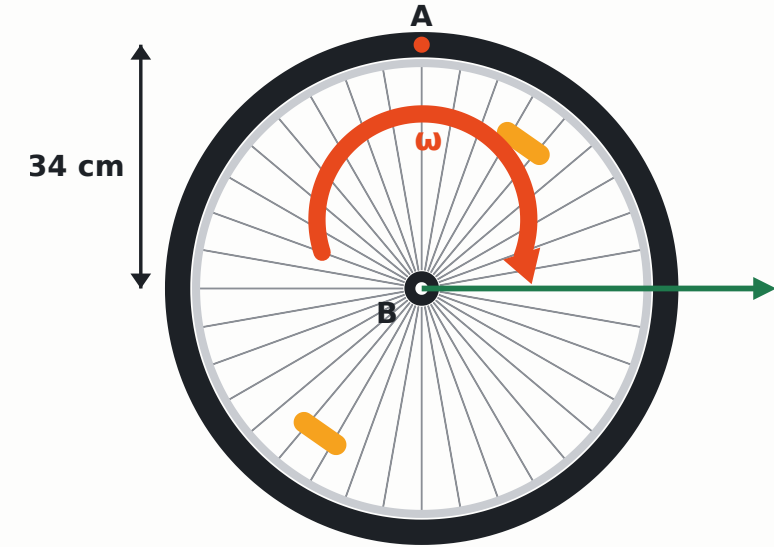
$$V_{A/G} = V_{A/B} + V_{B/G}$$

Then solve for $V_{A/B}$:

$$V_{A/B} = r\omega$$

$$V_{A/B} = (0.34 \text{ m})(2 \text{ rad/s})$$

$$V_{A/B} = \mathbf{0.68 \text{ m/s}}$$



Solution (continued)

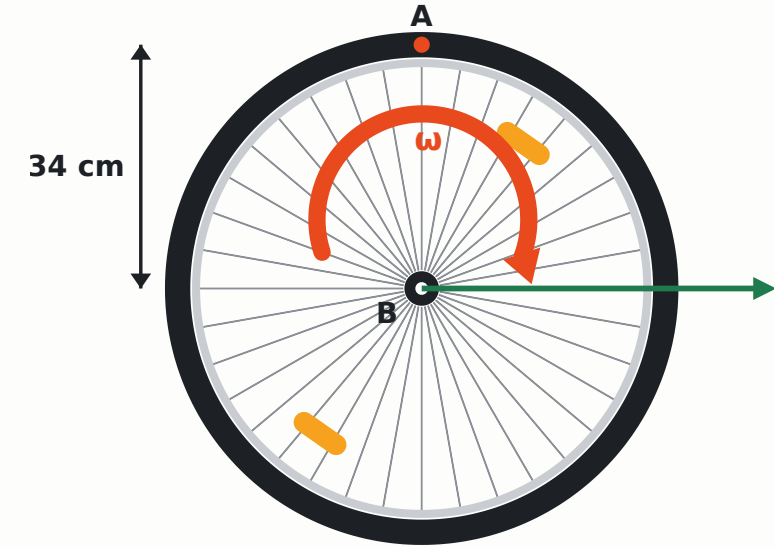
Then add the velocities to solve for $V_{A/G}$:

$$V_{A/G} = V_{A/B} + V_{B/G}$$

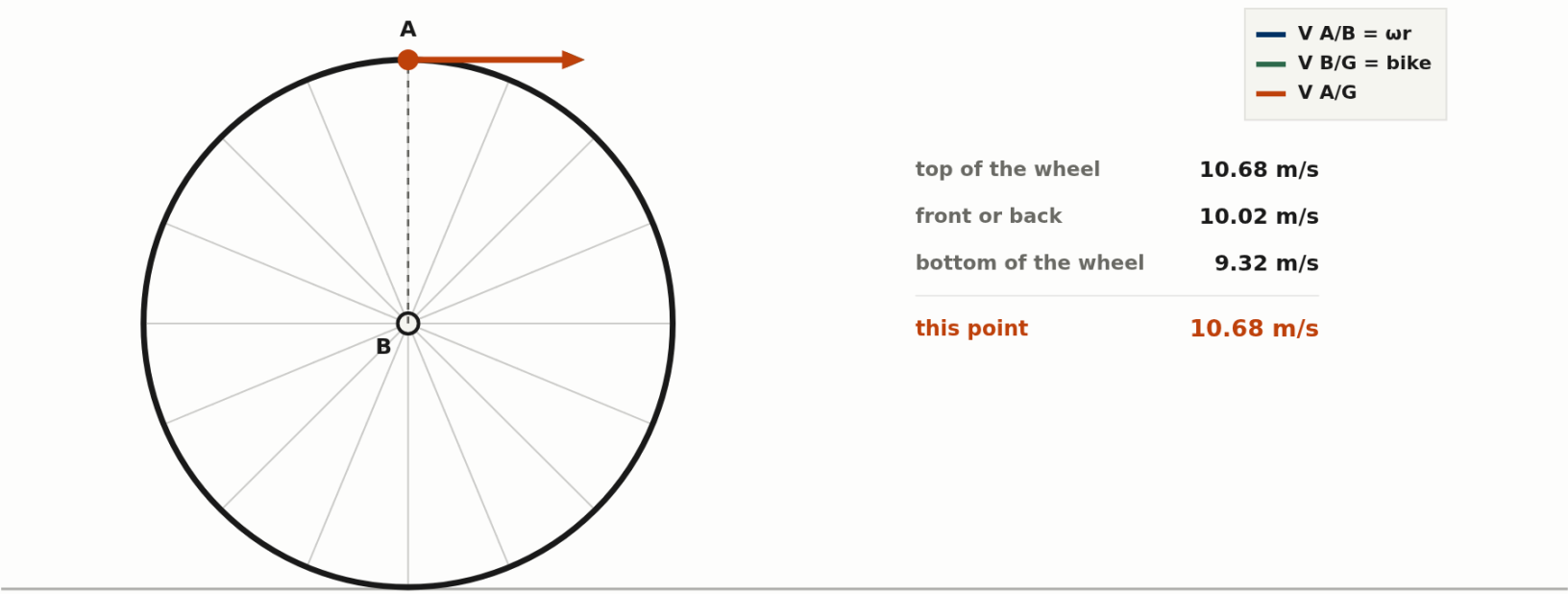
$$V_{A/G} = 0.68 \text{ m/s} + 10 \text{ m/s}$$

$$V_{A/G} = \mathbf{10.68 \text{ m/s}}$$

At the **top** of the wheel the tangential velocity points **forward**, the same way the bicycle is going — so the two simply add. No trigonometry is needed because both vectors lie along x.



Every point on the wheel at once



$\omega = 2.00 \text{ rad/s}$, so $V_{A/B} = \omega r = 0.68 \text{ m/s}$. At the top the two vectors line up and add to **10.68 m/s**; at the bottom they oppose and give **9.32 m/s**.

Point on the rim

90°

Bicycle velocity

10.0 m/s

Angular velocity ω

2.0 rad/s

$V_{B/G}$

Example Problem

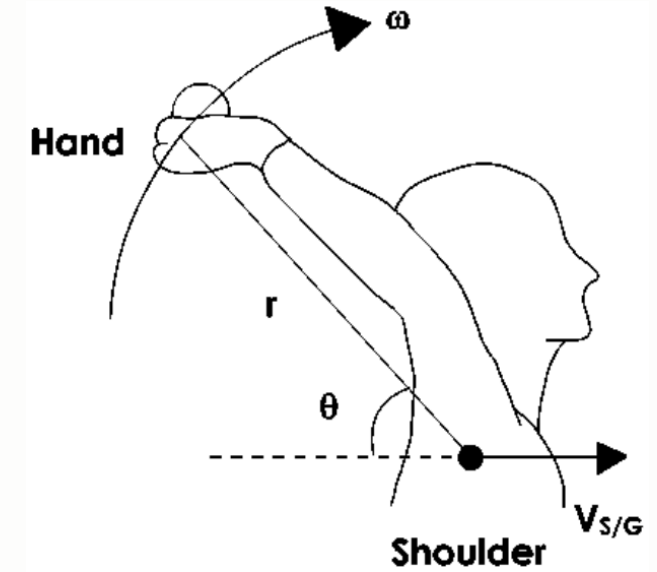
A baseball pitcher's shoulder is translating forward as his arm rotates about the shoulder. The objective of the motion is to deliver the required velocity (magnitude and direction) to the ball before it leaves his hand. Assume that the radial distance of the hand from the shoulder (r) is **0.58 m**, θ is **50 degrees** and ω is **4.2 rad/s**. Assume also that the shoulder has a velocity of **1.8 m/s** acting horizontally with respect to the ground. What is the velocity of the hand with respect to the ground (i.e. $V_{H/G}$)?



Draw a diagram

For simplicity, let us draw another diagram with all of the variables labelled on a **rigid segment representing the arm**.

The arm becomes one straight segment from shoulder to hand.
The shoulder is the moving joint B; the hand is the rotating point A.

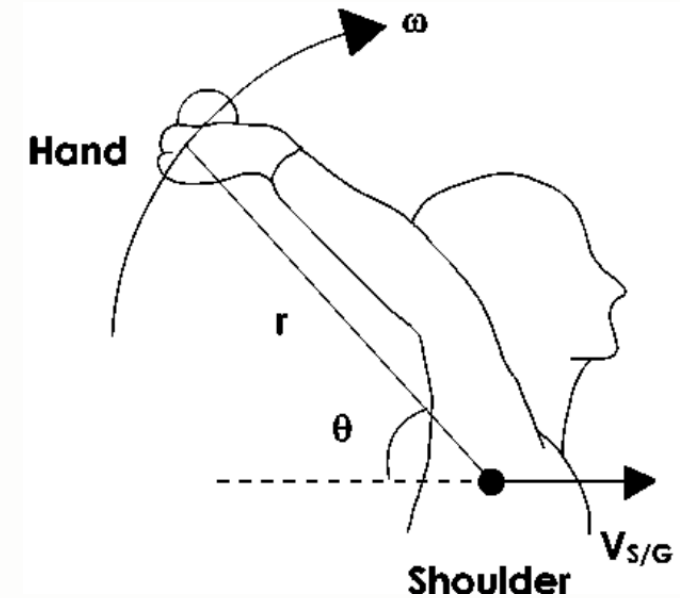


Develop an equation to describe the motion of the arm

The velocity is measured relative to the ground and is the vector sum of the ball's velocity relative to the shoulder and the shoulder relative to the ground.

The velocity of the Hand with respect to the ground ($V_{H/G}$) is equal to the velocity of the Hand with respect to the Shoulder ($V_{H/S}$) plus the velocity of the Shoulder with respect to the Ground ($V_{S/G}$).

$$V_{H/G} = V_{H/S} + V_{S/G}$$



Separate the equation into vertical (y) and horizontal (x) components

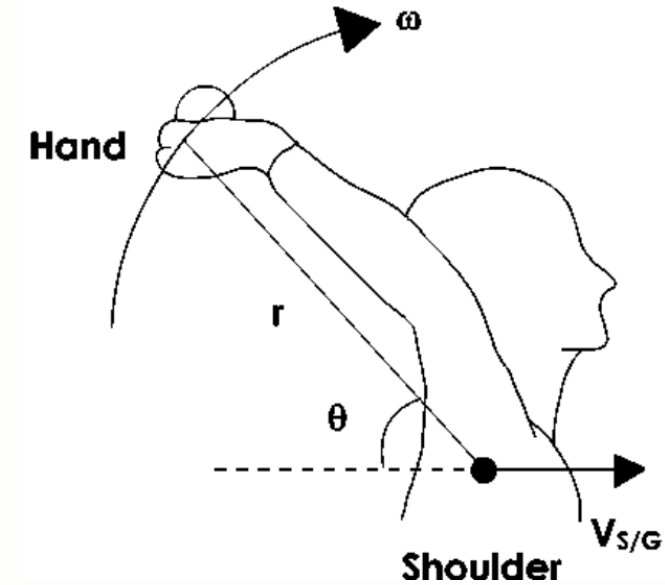
We also know that $V_{H/G}$ is equal to

$$V_{H/G} = V_{H/S} + V_{S/G}$$

Therefore it follows that

$$V_{H/Gx} = V_{H/Sx} + V_{S/Gx}$$

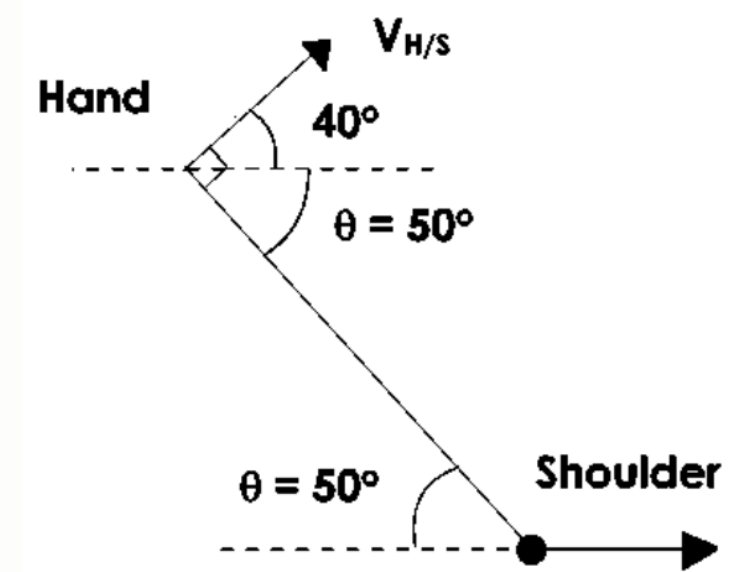
$$V_{H/Gy} = V_{H/Sy} + V_{S/Gy}$$



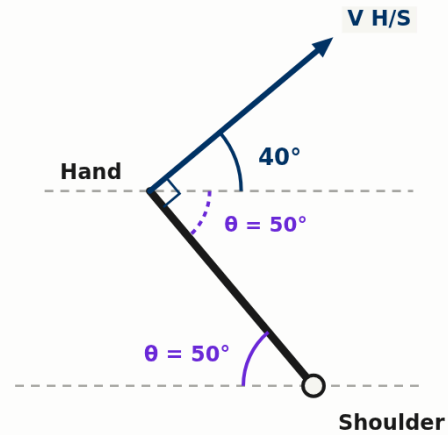
Separate the equation into vertical (y) and horizontal (x) components

- As you can see $V_{S/G}$ **only** has an x-component
- $V_{H/S}$ has **both x and y** components
- To determine the angle to use, draw a **dotted horizontal line through the hand**. It should now be obvious that the angle of $V_{H/S}$ with respect to the horizontal is **40°** because of the z-pattern (trig)

$\theta = 50^\circ$ is the angle of the *segment*. $V_{H/S}$ is perpendicular to the segment, so its angle to the horizontal is $90^\circ - 50^\circ = 40^\circ$.



Why 40° and not 50°



✓ $90^\circ - \theta = 40^\circ$

$V_x = r\omega \cos 40^\circ + 1.8 = 3.67$

$V_y = r\omega \sin 40^\circ = 1.57$

$V_{H/G} = 3.99 \text{ m/s at } 23.1^\circ$

✗ using $\theta = 50^\circ$

$V_x = r\omega \cos 50^\circ + 1.8 = 3.37$

$V_y = r\omega \sin 50^\circ = 1.87$

$V_{H/G} = 3.85 \text{ m/s at } 29.0^\circ$

The segment sits at **50°**; $V_{H/S}$ is perpendicular to it, so it sits at **40°** to the horizontal. Using θ by mistake gives **3.85 m/s** instead of **3.99 m/s**.

Segment angle θ ● **50°**

Angular velocity ω ● **4.2 rad/s**

Sub in known values and solve for component velocities

Now use the equations for the x and y components of $V_{H/G}$:

X component: $V_{H/Gx} = V_{H/Sx} + V_{S/Gx}$

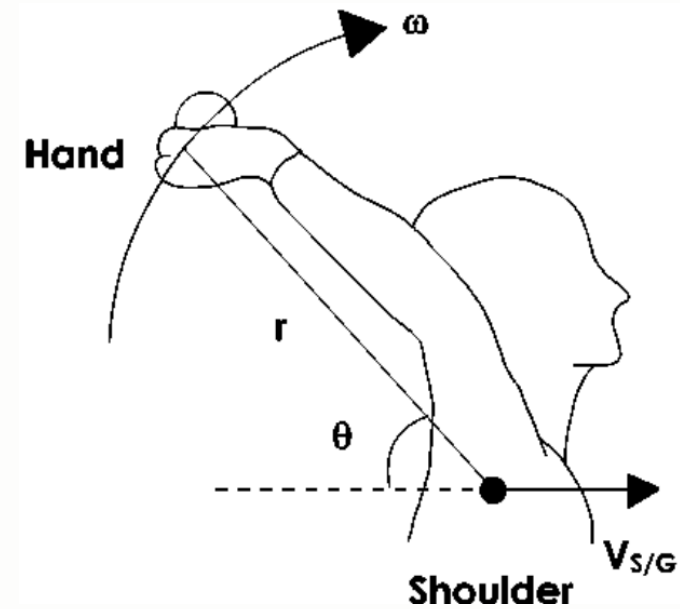
$$= r\omega \cos 40^\circ + 1.8$$

$$= \mathbf{3.67 \text{ m/s}}$$

Y component: $V_{H/Gy} = V_{H/Sy} + V_{S/Gy}$

$$= r\omega \sin 40^\circ + 0$$

$$= \mathbf{1.566 \text{ m/s}}$$



Solve for the resultant velocity

Now we need to solve for the **magnitude and direction** of $V_{H/G}$.

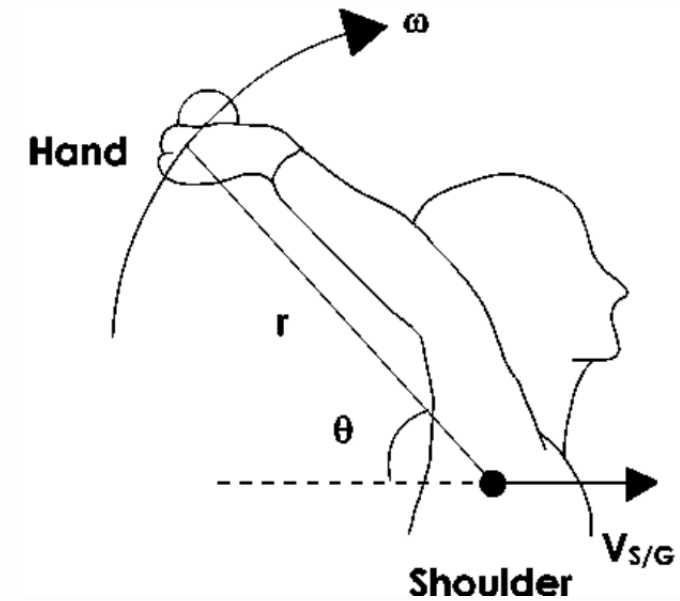
Magnitude:

$$V_{H/G}^2 = V_{H/Gx}^2 + V_{H/Gy}^2$$

$$V_{H/G} = \mathbf{3.99 \text{ m/s}}$$

Direction:

$$\tan^{-1}(1.566 / 3.67) = \mathbf{23.1^\circ}$$



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The diagram illustrates the kinematics of a human arm. A black line segment represents the arm, with the shoulder at the bottom and the hand at the top. The shoulder is labeled "Shoulder" and the hand is labeled "Hand". The arm segment is labeled $r = 0.58 \text{ m}$. The angle between the arm segment and a horizontal dashed line is $\theta = 50^\circ$. Two velocity vectors are shown: a blue vector $V_{H/S} = 2.44$ and an orange vector $V_{H/G}$. A green vector $V_{S/G} = 1.8$ is shown at the shoulder, pointing horizontally to the right. A right-angle symbol is shown at the hand, indicating that the velocity vector $V_{H/S}$ is perpendicular to the arm segment.

V H/G = 3.99 m/s at 23.1°

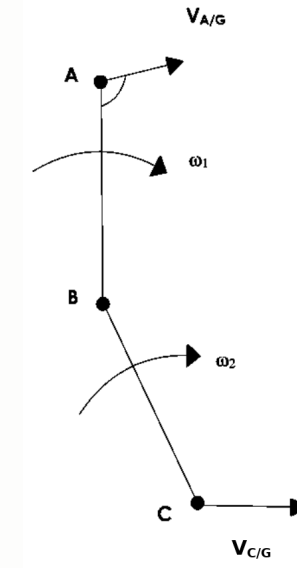
Segment angle θ 50°

Shoulder $V_{S/G}$ 1.8 m/s

... but what if we found that B was connected to another joint?

If we found that B was rotating about **another joint centre (C)** we would have to expand our equation to account for the tangential velocity of B about C, and no longer require the velocity of B relative to the ground:

$$V_{A/G} = V_{A/B} + V_{B/C} + V_{C/G}$$



Where...

$V_{C/G}$ is the linear velocity of C with respect to the ground

$V_{B/C}$ is the tangential velocity of B rotating about C

$V_{A/G}$ is the linear velocity of A with respect to the ground

$V_{A/B}$ is the tangential velocity of A rotating about B

Two of the four are **tangential** — one per rotating segment — and each is ωr for *its own* segment, with its own ω and its own r . Only the last term, the one touching the ground, is a plain linear velocity.

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The diagram shows a two-link robotic arm with joints at points B and A. Point C is the base of the first link, point B is the joint between the first and second links, and point A is the end of the second link. A horizontal ground line is shown below point C. Velocity vectors are shown at each point: $V_{C/G}$ (green, horizontal), $V_{B/C}$ (purple, perpendicular to link BC), $V_{A/B}$ (blue, perpendicular to link AB), and $V_{A/G}$ (orange, resultant velocity of point A). A legend in the top right corner provides the magnitudes: $V_{C/G} = 1.50$, $V_{B/C} = \omega_1 r_1 = 1.35$, $V_{A/B} = \omega_2 r_2 = 1.60$, and $V_{A/G} = 1.90 \text{ m/s}$.

$V_{C/G} = 1.50$
 $V_{B/C} = \omega_1 r_1 = 1.35$
 $V_{A/B} = \omega_2 r_2 = 1.60$
 $V_{A/G} = 1.90 \text{ m/s}$

$V_{A/G} = V_{A/B} + V_{B/C} + V_{C/G}$
1.90 m/s at 111.2° — $1.3\times$ the speed of the base

Angle of $C \rightarrow B$ **70°**

ω_1 (segment $C \rightarrow B$) **3.0 rad/s**

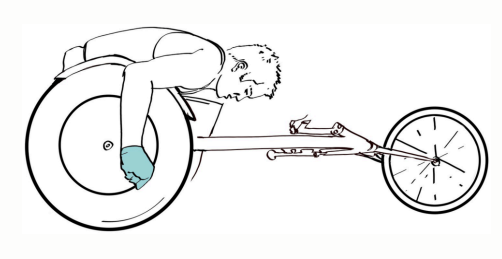
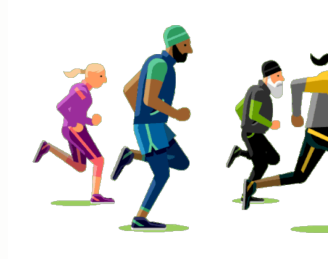
Angle of $B \rightarrow A$ **35°**

ω_2 (segment $B \rightarrow A$) **4.0 rad/s**

Why this matters

- Understanding general kinematics is important in many scenarios related to **human locomotion**
- We are often wanting to track the movement of the **whole body** and the **segments' velocity and displacement**

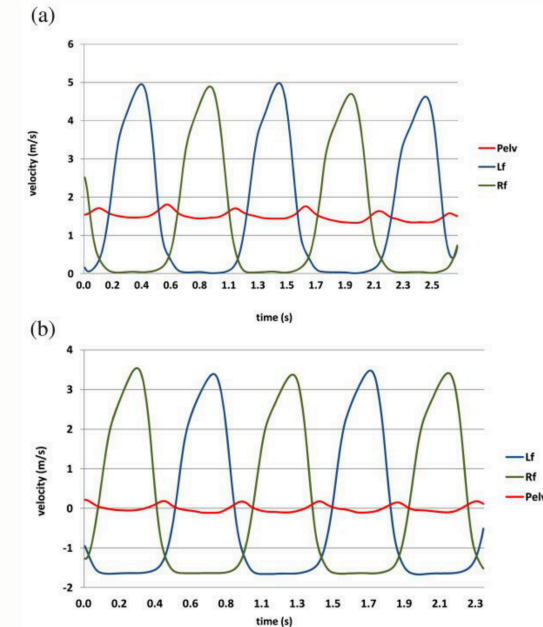
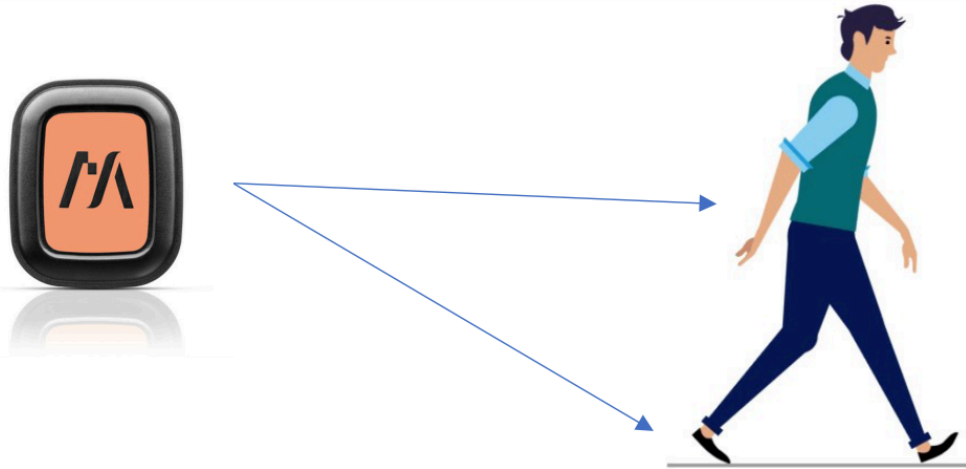
Walking, running, wheeling, throwing — in every one of them the joint you are rotating about is itself on the move.



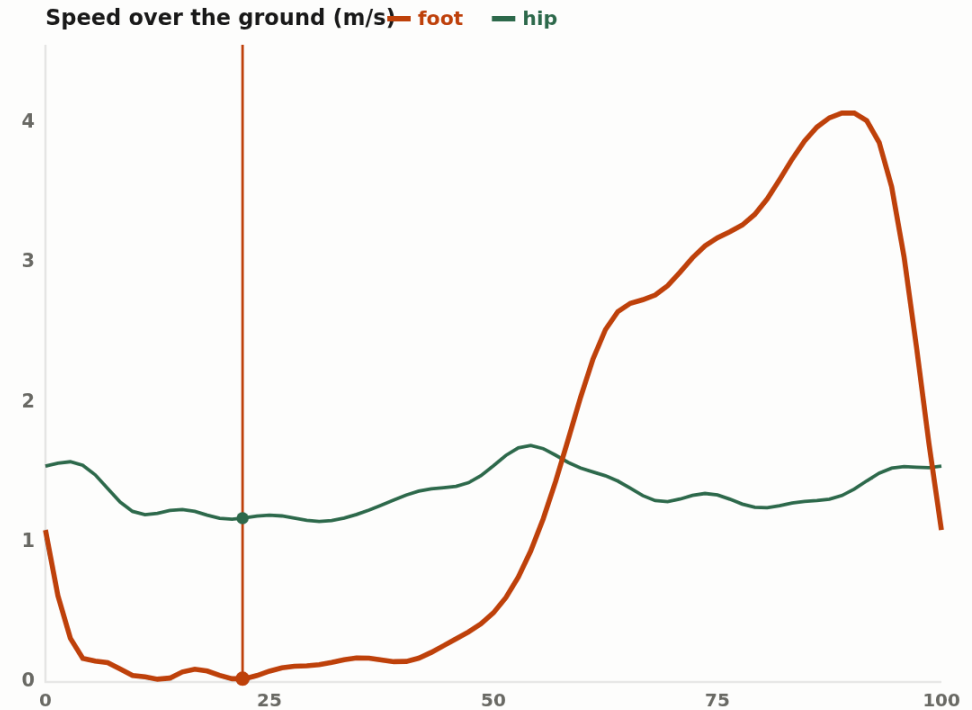
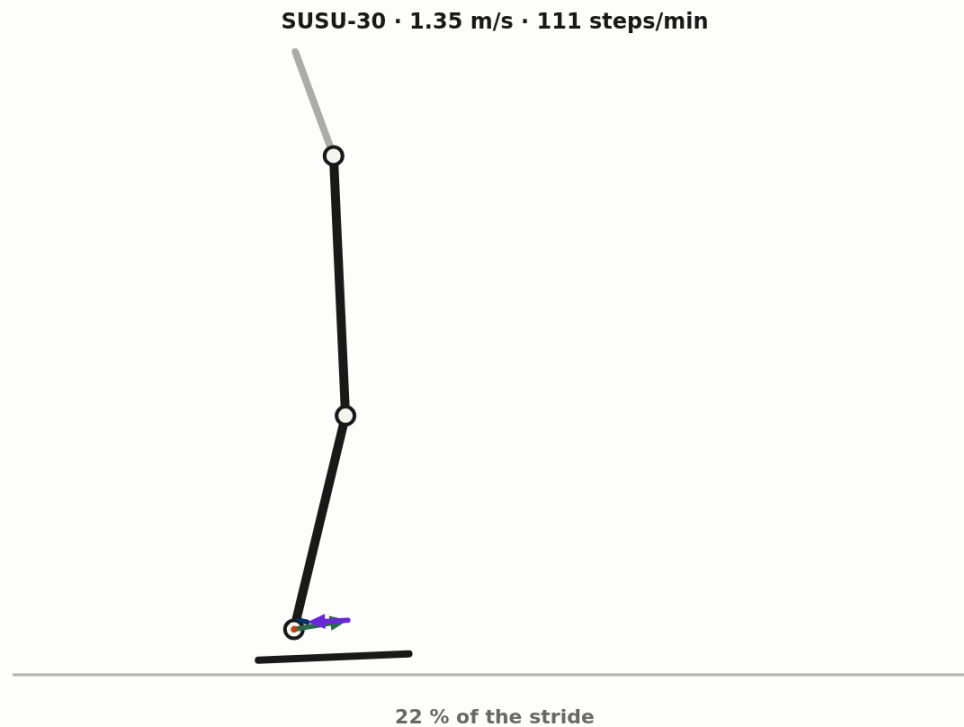
Feet, and what their velocity tells us

- Understanding the velocity and displacement of the **feet during walking** in relation to the whole body can give us insight into differences in walking gait related to
 - injury
 - aging
 - sex differences

We could use one IMU on each of the pelvis, thigh and shank ... **how could we get foot velocity?**



One real stride, built from the chain



At **22 %** of the stride the hip is doing **1.17 m/s** and the foot **0.02 m/s**. Over the stride the foot ranges from almost nothing on the ground to **4.07 m/s** in swing — **3.0×** walking speed. · The three terms reproduce the measured foot velocity to **0.11 m/s** on average, about 3 % of the peak — the residual is the rigid-segment assumption

Point in the stride  22 %

All three terms

Hip only

▶ Walk

What to take away

- **Velocity is always relative to something.** Name the frame before you give a number
- The rotating term is **tangential**: size ωr , direction perpendicular to the segment
- The angle of $V_{A/B}$ to the horizontal is **$90^\circ - \theta$** , not θ
- **$V_{A/G} = V_{A/B} + V_{B/G}$** — inner subscripts cancel, outer ones survive
- **Resolve, add, then recombine.** Components add; magnitudes do not
- **Chain it** for as many joints as you have — each rotating segment adds its own ωr

BIOMECHANICS (EPHE 341)

General kinematics

Next up: muscle mechanics · Further
reading: McGinnis Chpt 7